

1 Introduction

Question: What is Calculus?

Answer: Calculus is the Mathematics of change.

We may ask: How fast is a car moving at this instant? How much water has flowed into a tank up to this time? How does the temperature of a room change? How do engineers optimize the shape of a bridge or an aircraft wing? These are all questions involving continuous change. Whenever something changes continuously, calculus provides the mathematical language to study and analyze it.

Where is Calculus Used?

Calculus finds applications in almost every branch of science and engineering. Some examples include:

- Mechanical Engineering: velocity, acceleration, and the motion of machines.
- Electrical Engineering: current and voltage varying with time, and signal processing.
- Computer Science: optimization, machine learning, and neural networks.
- Civil Engineering: beam deflection and structural analysis.
- Chemical Engineering: reaction rates and diffusion.
- Data Science: optimization algorithms and error minimization.

These are only a few examples; calculus is a fundamental tool in many other disciplines as well.

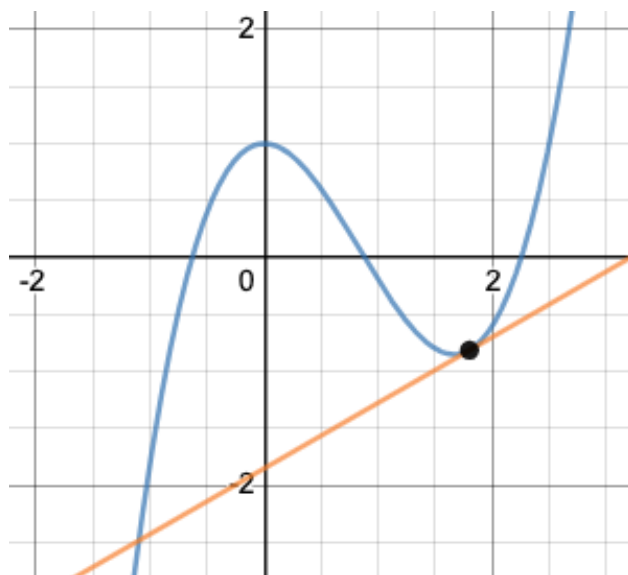
Calculus is a branch of mathematics concerned with the study of sequences, functions, limits, continuity, derivatives, integrals, and infinite series. It provides the mathematical language for understanding and analyzing continuous change.

Sir Isaac Newton (1642–1727), an English mathematician, physicist, and astronomer, and Gottfried Wilhelm Leibniz (1646–1716), a German mathematician, philosopher, and scientist, independently developed the foundations of calculus during the late seventeenth century.

1.1 Syllabus

The syllabus of the course is as follows. The course code is MA1110, and in this course, we will mainly discuss these three topics:

1. Sequence of real numbers: $\{a_1, a_2, \dots\}$.
2. Series of real numbers: $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \dots$.
3. Differential calculus, where we will discuss continuity and differentiability.



1. Sequence of Real Numbers.

By a sequence of real numbers, we mean an enumerated infinite collection of real numbers indexed by the set of natural numbers. Here, we will study the notion of limits for a sequence. If a sequence has a limit, then we call such a sequence a convergent sequence. We will study various properties of convergent sequences, and we will see many examples. Next, we study monotone sequences, that is, either monotone increasing or monotone decreasing sequences. At the end of this section, we give a criterion called the Cauchy criterion, which gives a necessary and sufficient condition for a sequence to be convergent without having any knowledge of the limit.

2. Series of Real Numbers.

In the second section, we discuss a series of real numbers. By a series of real numbers, we roughly mean adding infinitely many real numbers. The study of series is a major part of calculus. Series are used in most areas of mathematics, even for studying finite structures such as combinatorics through generating functions. Infinite series are also widely used in other disciplines such as physics, computer science, statistics, and finance.

For a long time, the idea that such a potentially infinite summation could produce a finite result was considered paradoxical. This paradox was resolved using the concept of a limit during the 17th century. This infinite summation can have a finite value—in this case, the series is called a convergent series; otherwise, it is called a divergent series. We will study various tests to determine whether a series is convergent or divergent.

3. Differential Calculus.

In the last part of this course, we will study differential calculus, which is a subfield of calculus that studies the rates at which quantities change. It is one of the two traditional divisions of calculus, the other being integral calculus, which is the study of the area beneath a curve. The primary objects of study in differential calculus are the derivative of a function and related notions. The derivative of a function at a chosen point describes the rate of change of the function near that point. Geometrically, this is the tangent at this point of this curve, and the derivative is given by the slope of this tangent line.

Differential calculus and integral calculus are connected by the Fundamental Theorem of Calculus, which states that differentiation is the reverse process to integration. Differentiation has applications to nearly all quantitative disciplines. For example, in physics, the derivative of the momentum of a body with respect to time equals the force applied to the body. Rearranging this statement leads to the famous equation $F = ma$ associated with Newton's second law of motion. In operations research, derivatives determine the most efficient ways to transport materials and design factories. Derivatives are frequently used to find the maxima and minima of a function. Equations involving derivatives are called differential equations and are fundamental in describing natural phenomena. Derivatives and their generalizations appear in many fields of mathematics, such as complex analysis, functional analysis, differential geometry, measure theory, and abstract algebra.

So, in this course, we will discuss mainly continuity and differentiability of a function of a single variable. We will study the rigorous definition of these two notions, and statements of Rolle's theorem, Lagrange's mean value theorem, and its applications.

1.2 Basic properties of the real numbers

Throughout this course, our focus will be on understanding not only how to solve problems, but also why the fundamental results of calculus are true. This requires a careful study of the real number system, which provides the foundation for the notions of limits, continuity, and differentiability.

Does every bounded sequence converge?

$$(-1)^n$$

Does every bounded increasing sequence converge?

Answer: Yes.

We know the statement is true, but mathematics asks **why it is true**. This naturally motivates the study of the real numbers. Before studying sequences and series, we must understand the number system in which they live. The most important property of the real numbers is **completeness**. Today, we begin with the basic properties of the real numbers.

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^c \subset \mathbb{C}$$

The students are expected to be familiar with the algebraic operations on the real numbers, namely addition, additive inverse, multiplication, and multiplicative inverse. They may also wish to briefly revisit the basic notions of sets, relations, binary relations, partially ordered sets, and totally ordered sets. In this course, we shall begin with the fact that $(\mathbb{R}, \leq)$ is a totally ordered set. One of the distinguishing features of the real numbers is that any two real numbers can be compared. The first fundamental property of this order relation is the

Law of Trichotomy

The Law of Trichotomy states that for any two real numbers x and y , exactly one of the following statements holds:

1. $x = y$;
2. $x < y$;
3. $y < x$.

1.2.1 Upper and lower bounds

Definition 1. Let $A \subset \mathbb{R}$ be nonempty. A real number α is called an upper bound of A if for each $x \in A$, we have $x \leq \alpha$.

$$\text{Quantifiers:} \quad \forall x \in A (x \leq \alpha).$$

Geometrically, this means that elements of A are to the left of α on the number line.



Figure 1: α is an upper bound of A

A real number α is not an upper bound of A if there exists at least one $x \in A$ such that $x > \alpha$. In terms of quantifiers

$$\exists x \in A (x > \alpha).$$

Geometrically, this means that we can find an element of A to the right of α on the number line.

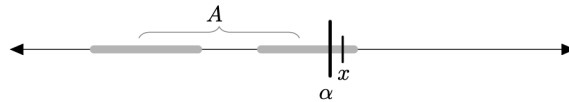


Figure 2: α is not an upper bound of A

If α is an upper bound of A and $\alpha' > \alpha$, then α' is also an upper bound of A .

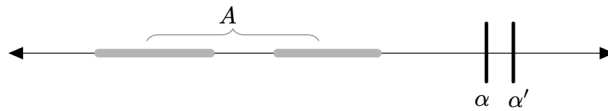


Figure 3: Upper bound of a set is not unique

Exercise 1. Define the notion of a lower bound for a nonempty subset of $\mathbb{R}$.

Exercise 2. Show that $\mathbb{N}$ has a lower bound in $\mathbb{R}$. Does $\mathbb{N}$ have an upper bound in $\mathbb{R}$?

Definition 2. Let $A \subset \mathbb{R}$ be nonempty. We say that A is bounded above in $\mathbb{R}$ if there exists an upper bound of A . Equivalently, there exists $\alpha \in \mathbb{R}$ such that

$$x \leq \alpha, \quad \forall x \in A.$$

In terms of quantifiers, the above definition may be written as

$$\exists \alpha \in \mathbb{R} (\forall x \in A (x \leq \alpha)).$$

Definition 3. Let $A \subset \mathbb{R}$ be nonempty. We say that A is not bounded above in $\mathbb{R}$ if for every $\alpha \in \mathbb{R}$, there exists $x \in A$ (which may depend on α) such that $x > \alpha$.



Figure 4: Set which is not bounded above

In terms of quantifiers, the above definition may be written as

$$\forall \alpha \in \mathbb{R} (\exists x \in A (x > \alpha)).$$

Exercise 3. Using the definition of bounded above, define what it means for a subset of $\mathbb{R}$ to be bounded below. Can you visualize this on the number line? When do you say that a subset $A \subset \mathbb{R}$ is not bounded below in $\mathbb{R}$?

An upper bound of a set need not be an element of the set. For example, if

$$A = (a, b),$$

then b is an upper bound of A , but $b \notin A$. If an upper bound of A is itself an element of A , then it is called the *maximum* of A .

Exercise 4. Suppose that $M, M' \in A$ are both maximum of a nonempty subset $A \subset \mathbb{R}$. Prove that $M = M'$.

Exercise 5. Define the minimum of a nonempty subset $A \subset \mathbb{R}$. Prove that if A has a minimum, then it is unique.

Theorem 1. Let $A \subset \mathbb{R}$ be a nonempty set. If α is a lower bound of A and β is an upper bound of A , then

$$\alpha \leq \beta.$$

Proof. Since A is nonempty, there exists an element $a \in A$. As α is a lower bound of A , we have

$$\alpha \leq a.$$

Similarly, since β is an upper bound of A , we have

$$a \leq \beta.$$

Hence,

$$\alpha \leq a \leq \beta,$$

and therefore

$$\alpha \leq \beta.$$

□

Definition 4. A subset $A \subseteq \mathbb{R}$ is said to be bounded if it is both bounded above and bounded below. Equivalently, there exist real numbers m and M such that $m \leq x \leq M$ for every $x \in A$.

Definition 5. Let $A \subset \mathbb{R}$ be a nonempty set that is bounded above. A real number $\alpha \in \mathbb{R}$ is called the least upper bound (LUB) or supremum of A if

1. α is an upper bound of A ; and
2. if β is any upper bound of A , then $\alpha \leq \beta$.

Notationally we write: $\alpha = \sup A$ or $\alpha = \text{lub } A$.

Definition 6. Let $A \subset \mathbb{R}$ be a nonempty set that is bounded below. A real number $\alpha \in \mathbb{R}$ is called the greatest lower bound (GLB) or infimum of A if

1. α is a lower bound of A ; and
2. if β is any lower bound of A , then $\beta \leq \alpha$.

Notationally we write: $\alpha = \inf A$ or $\alpha = \text{glb } A$.

Example 1. Let

$$A = (0, 1).$$

Then

$$\sup A = 1, \quad \inf A = 0.$$

Notice that neither 0 nor 1 belongs to A .

Example 2. Let

$$A = [0, 1].$$

Then

$$\sup A = 1, \quad \inf A = 0.$$

In this case,

$$\max A = 1, \quad \min A = 0.$$

Thus, the supremum and infimum belong to the set.

Example 3. *Let*

$$A = (0, 1].$$

Then

$$\sup A = 1, \quad \inf A = 0.$$

Moreover,

$$\max A = 1,$$

but A has no minimum since $0 \notin A$.

Example 4. *Let*

$$A = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}.$$

Then

$$\sup A = 1, \quad \inf A = 0.$$

Moreover,

$$\max A = 1,$$

but A has no minimum.

Example 5. *Let*

$$A = \mathbb{N}.$$

Then A is bounded below but not bounded above. Here, $\inf A = 1$, whereas $\sup A$ does not exist in $\mathbb{R}$.

Example 6. *Let*

$$A = \{x \in \mathbb{R} : x^2 < 4\}.$$

Since

$$A = (-2, 2),$$

we have

$$\sup A = 2, \quad \inf A = -2.$$

However, A has neither a maximum nor a minimum.